

Independent Computational Verification

Paper II — Section 3: the vertex-copy inequality (Lemma 3.1)

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Artifact: A_vertex_copy_inequality/audit_A_vertex_copy.py **Verdict:** PASS

Scope

Exhaustive over **all non-isomorphic graphs on 2–7 vertices** (NetworkX atlas), and for **every** nonadjacent vertex pair (u, v) in each graph.

Claims verified

For nonadjacent u, v , writing $G_{v \rightarrow u}$ for the copy operation:

- **(I) Lemma 3.1:** $\Phi_\tau(G_{v \rightarrow u}) + \Phi_\tau(G_{u \rightarrow v}) \geq 2\Phi_\tau(G)$.
- **(II) eq. 3.4:** $e(G_{v \rightarrow u}) + e(G_{u \rightarrow v}) = 2e(G)$ (exact).
- **(III) eq. 3.3:** $\tau_3^*(G_{v \rightarrow u}) + \tau_3^*(G_{u \rightarrow v}) \leq 2\tau_3^*(G)$.
- **Cross-check:** $\tau_3^*(G) = \nu_3^*(G)$ (LP duality), computed by two independent LPs.

Method

Python 3, NetworkX (graph atlas + chordality), SciPy `linprog` method=`highs`; exact fractions where noted. $\Phi_\tau(G) = |E(G)| - 2\tau_3^*(G)$, with τ_3^* the optimum of the triangle-cover LP.

Results

n	non-iso graphs	nonadjacent pairs tested
2	2	1
3	4	6
4	11	33
5	34	170
6	156	1170
7	1044	10962

- Graphs tested: **1251**; nonadjacent pairs tested: **12342**; max n : **7**.
- Minimum slack of (I): **-7.11e-15** (numerically zero; inequality holds, attained with equality).
- Max absolute error of identity (II): **0.00e+00** (exact).
- Max violation of (III): **0.00e+00** (none).
- Max LP duality gap $|\tau_3^* - \nu_3^*|$: **2.66e-15**.
- **Violations:** **0**. Runtime 142.24 s.

Verdict

PASS. The vertex-copy inequality, both exact identities, and LP duality hold on the entire enumerated universe with no violation. (Values at the 10^{-15} level are floating-point noise.) This is the same *L1* step machine-checked in the Lean development.

*This document reports an **independent computational verification** of the stated claims: each result was recomputed from scratch by a self-contained script (exact linear programming via SciPy/HiGHS for the fractional triangle-cover number, exact rational arithmetic for the integer optimization, and exhaustive enumeration of non-isomorphic graphs via the NetworkX atlas where indicated). This is computational evidence corroborating the analytic proof; it is not itself a formal proof. The machine-checked formal proof is the Lean development, which is conditional on the classical chordal-structure theory.*